

| Question | Answer                                          | Mark                   | Mark scheme                                                                                                                                                                                                                                                    | Additional guidance                                    |
|----------|-------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 16 (a)   | $x_1 = 1.817$<br>$x_2 = 1.853$<br>$x_3 = 1.846$ | M1<br><br>M1<br><br>A1 | for a correct method to find $x_1$ eg $\sqrt[3]{10 - 2 \times 2}$ (= 1.8171.....)<br><br>(dep on M1) for substitution of $x_1$ to give $x_2$ <b>and</b> $x_2$ to give $x_3$<br><br>for $x_1 = 1.81(71\dots)$ , $x_2 = 1.85(33\dots)$ and $x_3 = 1.84(62\dots)$ | Accept an accuracy of 2dp or more rounded or truncated |
| (b)      | $a = 2$ , $b = -10$                             | C1                     | cao                                                                                                                                                                                                                                                            |                                                        |